Problem Set 1

Due date: 11:59pm, 9/12/2025

This problem set is divided into two portions: first, an optional review of prerequisites to this class (problems marked "review" in bold), followed by problems on the new material we have discussed in lecture. The review problems are optional, but those who do all of them completely correctly will receive **10 extra credit points**. No partial credit will be given for the review problems. I strongly urge you to do the review problems, as I will assume that you know all of those results in class.

1. **Review: curves and tangents in $\mathbb{R}^3$:** Consider the following parametric equation in Cartesian coordinates:

$$\mathbf{v}(\lambda) = \big(\sin(\lambda^2), \ln(\lambda+1), 2\lambda^2 - 1\big)$$

   a. Compute the tangent vector to this curve at any value of λ.
   b. Draw a 3D plot for $\mathbf{v}(\lambda)$ from $\lambda = 0$ to $\lambda = 1$. You may use computer software to draw this if you wish. *By hand*, draw the tangent vectors to $\mathbf{v}(\lambda)$ at $\lambda = 0$, $\lambda = 1/2$, and $\lambda = 1$.
   c. Transform $\mathbf{v}(\lambda)$ to spherical coordinates in the range $0 < \lambda < 1$.
   d. Compute $\mathbf{v}(\lambda) \cdot \mathbf{v}(\lambda)$.
   e. Compute $\mathbf{v}(\lambda) \times \mathbf{v}'(\lambda)$ in your choice of coordinates.

2. **Review: Short Problems:** As always, make sure you show your work (and do not use software) to get credit.

   a. **Chain rule:** let $f(x(\lambda), y(\lambda), \lambda) = \lambda^2 + \sin(x(\lambda)) + \frac{y(\lambda)}{x(\lambda)}$. Compute $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ and $\frac{df}{d\lambda}$.
   b. **Taylor expansion** Taylor expand the following expression around $x = 0$ up to and including $\mathcal{O}(x^2)$: $\frac{1}{(1-x^2)^{1/3}}$.
   c. **Differential equation:** Solve the following differential equation:

$$f''(x) - 2f(x) = 0$$

   and write your solution in terms of the hyperbolic trig functions sinh and cosh.

3. **Review: Rotation Matrix:** Given a point (x_0, y_0, z_0) in Cartesian coordinates in $\mathbb{R}^3$, we would like to rotate that point an angle θ about the origin in the $x - y$ plane.

   a. Write down the new coordinates of the point after this rotation.
   b. Show that the relation can be written as a linear algebra equation $X' = R[\theta] \cdot X$ where $X = (x_0, y_0, z_0)$, X' is the vector you obtained in part (a), and $R[\theta]$ is a matrix that you will need to write down.

4. **Review: Matrix Multiplication:** Given two square matrices $[M]$ and $[N]$:

$$M = \begin{pmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{pmatrix} \qquad N = \begin{pmatrix} N_{11} & N_{12} & N_{13} \\ N_{21} & N_{22} & N_{23} \\ N_{31} & N_{32} & N_{33} \end{pmatrix}$$

a row vector

$$v = \begin{pmatrix} v_1 & v_2 & v_3 \end{pmatrix}$$

and a column vector

$$u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$$

write down the following quantities in terms of their entries, using the Σ sum notation.

a. $[M] \cdot [N]$.

b. $[N] \cdot [M]$.

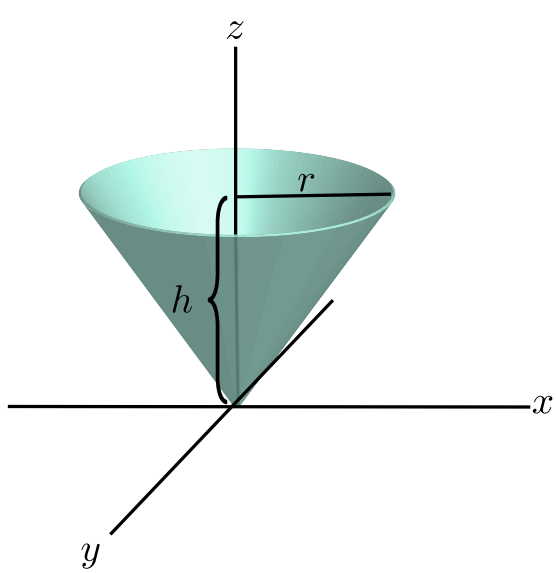
c. $v \cdot [M] \cdot u$.

d. $v \cdot u$.

5. *(20 points)* Using Galilean relativity, show using equations that both Newton's second law $\mathbf{F} = m\mathbf{a}$ and Newton's law of gravitation $\mathbf{F} = -G\frac{m_1 m_2}{r^2}\hat{\mathbf{r}}$ are consistent with the Principle of Relativity (that is, that reference frames with constant relative velocity between them agree on both laws). For this you need only transform $m\mathbf{a}$ and $-G\frac{m_1 m_2}{r^2}\hat{\mathbf{r}}$.

6. *(15 points)* Explain why it makes sense to talk about observer 1 being "at rest relative to" observer 2 or "moving at some velocity" relative to observer 2, but it does *not* make sense to talk about observer 1 being "inertial relative to" observer 2.

7. *(25 points)* Consider an upwards-pointing two-dimensional cone with radius r and height h:



a. *(5 points)* Write down the parametric equations for the cone in $\mathbb{R}^3$ using h and r and Cartesian

coordinates. That is, write down the equations of the form:

$$x =$$
$$y =$$
$$z =$$

In terms of some parameters (I recommend θ and z). Be sure to specify the range of those parameters.

b. *(5 points)* Write down both the equations for the cone in terms of the opening angle, call it α, rather than h and r. Note that since these equations do not invoke h and r, you can use them to specify an infinitely-extended cone. What range of parameters does that correspond to?

c. *(15 points)* Write down the equations for the *solid* (three-dimensional) cone (including its boundary) in terms of α and your parameters. Use this to prove (using equations) that any point p inside the solid cone lies on a line from the origin to p, where the angle that this line makes with the z-axis is smaller than $\alpha/2$. Conversely, show (using either equations or no more than two sentences in words) that any point that lies on such a line must live inside of the solid cone.

8. *(20 points)* Let Alice and Bob both be inertial observers. Bob is moving relative to Alice at some v_{BA} in the $+x$ direction. At $x_A = 0$, Bob is at some $t_A = t_0 > 0$ (the A subscripts refer to coordinates in Alice's rest frame). Draw Bob's worldline in the spacetime diagram in Alice's frame. Charlie is moving relative to Alice in the $+x$ direction at some $v_{CA} > v_{BA}$. At $x_A = 0$, Charlie is at $t_A = 0$. Draw Charlie's worldline in the same spacetime diagram. Show using equations that in computing Bob's velocity relative to Charlie, the time offset is immaterial.

9. *(40 points)* Recall that an event B lies in the future of an event A if and only if it lies in the solid cone bounded by light rays. Using the fact that light rays travel at 45 degree angles and the results of problem 7(c), prove that B lies in the future of A if and only if there there is a single (sublight) worldline that passes through A and then through B. *Note that you are being asked to prove an if and only if statement: that means you must prove that if B lies in the future of A, then there is a sublight worldline that passes through A and then through B AND also prove that if there is a sublight worldline that passes through A and then through B then B lies in the future of A.*

10. *(20 points)* Bob and Alice are inertial observers with a relative velocity of $0.5c$ between them. Their worldlines cross at your origin. At this event, Alice briefly turns on a flashlight, creating a short burst of light.

a. *(5 points)* Draw a spacetime diagram in Alice's reference frame, assuming that Alice perceives Bob to be moving in her $+x$ direction. Write down the angle that Bob's worldline makes with Alice's t-axis.

b. *5 point* Let events A and B be the events on the left and right light rays (respectively) that are one second (as determined by Alice's reference frame) after the burst of the light. Label these two events on your diagram.

c. *(10 points)* Let event C be the event on the left light ray that is simultaneous with B as determined by *Bob*. Mark this event on the diagram. Draw Bob's line of simultaneity going through event C.